3. Water Jug Problem

Aim:

To write a Python program to solve the Water Jug problem using a state-space search.

Algorithm:

1. Start the program.
2. Define the capacities of two water jugs.
3. Define the initial state as (0, 0).
4. Define the goal amount of water.
5. Generate possible operations: fill, empty, and pour.
6. Check each new state.
7. Repeat until the goal state is reached.
8. Display the solution.
9. Stop.

Code:

```
from collections import deque

def water_jug():
    q = deque([(0, 0)])
    visited = set()

    while q:
        a, b = q.popleft()

        if (a, b) in visited:
            continue

        visited.add((a, b))
        print(a, b)

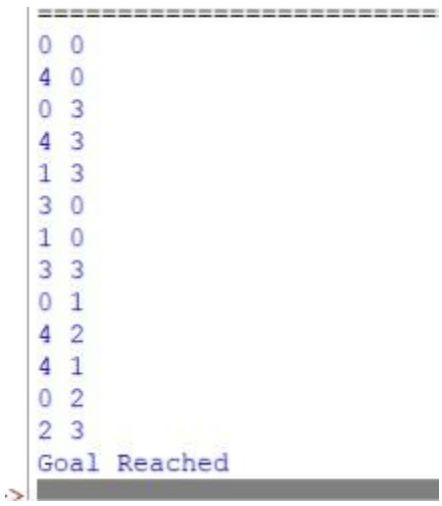
        if a == 2:
            print("Goal Reached")
            return

        states = [
            (4, b), (a, 3),
            (0, b), (a, 0),
            (a-min(a, 3-b), b+min(a, 3-b)),
            (a+min(b, 4-a), b-min(b, 4-a))
        ]

        for state in states:
            if state not in visited:
                q.append(state)

water_jug()
```

Output:



Result:

Thus, the Python program to solve the Water Jug problem was executed successfully.